

Machine Learning and Deep Learning Models for Dengue Diagnosis Prediction: A Systematic Review

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Abstract: The global crisis triggered by the dengue outbreak has increased mortality and placed significant pressure on healthcare services worldwide. In response to this crisis, there has been a notable increase in research employing machine learning and deep learning algorithms to anticipate diagnosis in patients with suspected dengue. To conduct a comprehensive systematic review, a detailed analysis was carried out to explore and examine the machine learning methodologies applied in diagnosing this disease. An exhaustive search was conducted across numerous scientific databases, including Scopus, IEEE Xplore, PubMed, ACM, ScienceDirect, Wiley, and Sage, encompassing studies up to May 2024. This extensive search yielded a total of 2723 relevant articles. Following a rigorous evaluation, 32 scientific studies were selected for the final review, meeting the established criteria. A comprehensive analysis of these studies revealed the implementation of 48 distinct machine learning and deep learning algorithms, showcasing the heterogeneity of methodological approaches employed in the research domain. The results indicated that, in terms of performance, the support vector machine (SVM) algorithm was the most efficient, being reported in 25% of the analyzed studies. The Random Forest algorithm was the second most frequently used, appearing in 15.62% of the 32 reviewed articles. The PCA-SVM algorithm (poly-5), a variant of SVM, emerged as the best-performing model, achieving 99.52% accuracy, 99.75% sensitivity, and 99.09% specificity. These findings offer significant insights into the potential of machine learning techniques in the early diagnosis of dengue, underscoring the necessity to persist in exploring and refining these methodologies to enhance clinical care in cases of this disease.

Keywords: algorithms; dengue diagnosis; deep learning; diagnostic model; machine learning; support vector machine



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1. Introduction

Dengue viral infection is a pandemic phenomenon transmitted through the bites of female mosquitoes of the *Aedes albopictus* or *Aedes aegypti* species [1]. The infection has been shown to cause a progressive deterioration in the health of infected individuals and poses a significant threat in terms of human morbidity and mortality [2].

Consequently, dengue has become a global health concern [3]. According to World Health Organization (WHO) estimates, approximately 50 to 100 million dengue infections

occur annually in over 100 endemic countries, placing nearly half of the global population at risk [4]. Of these cases, almost 75% occur in Asia and the Western Pacific region. This results in an annual incidence exceeding 40 million symptomatic episodes and more than 20,000 deaths [5].

In 2022, the Americas reported 2.8 million dengue cases, compared to 1.2 million in 2021, signifying an increase of more than double. According to data provided by the WHO, approximately three million dengue cases, both suspected and confirmed (2,997,097), were reported in the American regions as of 1 July 2023. This figure was associated with 1302 reported deaths, resulting in a mortality rate of 0.04%. Furthermore, an analysis of the data reveals that Brazil, Peru, and Bolivia have consistently exhibited more dengue cases than other countries in the region [6].

The WHO has classified dengue as a significant global concern due to its considerable impact on public health worldwide [7]. Projections indicate that by 2080, the global population at risk of contracting dengue will exceed 6 billion, representing a more than twofold increase compared to the number of cases reported in 2015 [8]. Projections indicate that approximately 4 billion individuals worldwide, constituting at least 50% of the global population, are likely to be affected by dengue. Furthermore, approximately 400 million infections are projected annually [6,9].

The absence of an effective vaccine to prevent dengue infections and the fact that the existing vaccine does not provide equal protection against all four serotypes of the virus are of particular concern [10,11]. Consequently, numerous health authorities have directed their efforts towards implementing early warning systems [12]. The application of diverse modeling approaches for the early and accurate prediction of dengue could not only mitigate the threat posed by the disease, but also empower policymakers to implement effective strategies for its control [10]. A global survey of early warning systems for dengue reveals the deployment of diverse modalities, with each nation selecting the most suitable approach for its specific context [13]. Examples include operational systems in South America, Malaysia, Singapore, and other regions.

In regions where dengue is endemic, particularly in Southeast Asia, the Americas, and the Pacific, conventional dengue surveillance predominantly relies on hospital syndromic reporting and laboratory confirmation [14,15]. While this conventional surveillance approach is renowned for its high accuracy, its constrained responsiveness engenders considerable delays in the identification and notification of cases [10]. The absence of integration between entomological and disease surveillance systems has been identified as a key factor that undermines the effectiveness of interventions, resulting in inaction or delays in implementing preventive and control measures for dengue [16]. The limited coverage of disease surveillance systems is evident in their failure to capture all cases, resulting in chronic underreporting and an inadequate representation of the actual morbidity burden in the region [14,15]. Integrating artificial intelligence-based systems and developing more robust approaches is essential to address these limitations effectively. In light of these considerations, the Special Program for Research and Training in Tropical Diseases (TDR) of the WHO, in collaboration with research institutions, national dengue control services, and academia in ten endemic countries, has undertaken the development of an Early Warning and Response System (EWARS) network for dengue [17].

As evidenced by previous studies, the EWARS has demonstrated adequate predictive performance in various settings [18]. This system is designed to be operated by trained and untrained users, which could have significant operational implications [17]. Nevertheless, the assessment of the impact of EWARS in endemic areas remains a crucial aspect of the comprehensive early warning and response process [19]. Conversely, numerous studies have explored innovative strategies based on Big Data [20] and machine learning and deep

learning algorithms to optimize the management of dengue outbreaks and other health-related topics [10,21,22]. However, recent systematic reviews on dengue have focused on analyzing epidemiological and clinical outcomes [15], surveillance systems [23], vaccine efficacy [24], the prevalence and burden of infection [25], clinical prognostic models [26], Bayesian spatial and spatiotemporal approaches [27], and the frequency of use of machine learning models in dengue [28] based on clinical data [29], meteorological data [10], climatic data [30], and demographic data [31]. However, these studies have not systematically reviewed the model's efficiency based on machine learning and deep learning. Consequently, they have not thoroughly discussed these novel methods.

The objective of the present study is to undertake a systematic review and thoroughly analyze the performance of various machine learning and deep learning algorithms. These algorithms are employed in creating predictive models for the early identification of dengue fever based on manifested symptoms.

2. Methodology

This research was developed using the systematic review methodology [32] based on the PRISMA guidelines (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [33,34].

The systematic review was conducted following a rigorous sequence of stages. The process began with the precise formulation of the research objective, which served as the guiding framework for the study. Consequently, a comprehensive research protocol was formulated, meticulously outlining the design of the systematic review. A list of the consulted databases was presented, and the search strategies employed for each were detailed. The inclusion and exclusion criteria were clearly established and de-scribed. The search and reference selection process was thoroughly illustrated. The characteristics of the included studies were precisely described. Finally, the results of the risk-of-bias assessment were presented, using the Quality Assessment of Diagnostic Accuracy Study 2 (QUADAS-2) tool to assess both the risk of bias and applicability concerns. The domains evaluated by QUADAS-2 include patient selection, the application and interpretation of the index test, the adequacy of the reference standard, patient flow and the timing between tests, and the clinical applicability of the analyzed studies. A presentation of the results, including the distribution of articles by country and continent.

2.1. Search Strategies

From 1 April to 30 May 2024, an exhaustive search was conducted across various academic databases. The following table details the search methodology. Predefined search terms were utilized, and the established selection criteria were meticulously applied. Table 1 summarizes the databases consulted, along with the number and percentage of documents retrieved.

Table 1. Databases consulted.

IDE	Database	Quantity	Percentage (%)
DB1	Scopus	386	14.18
DB2	IEEE Xplore	120	4.4
DB3	Pubmed	49	1.79
DB4	ACM	145	5.33
DB5	ScienceDirect	1433	52.63
DB6	Wiley	537	19.72
DB7	Sage Journals	53	1.95
	Total	2723	100

The researchers (DA, WR) thoroughly analyzed the titles and abstracts of the identified articles, prioritizing those that employed data based on clinical diagnoses, demographic information, climate variables, meteorological data, and/or images related to environmental density. Furthermore, the researchers verified the strict adherence to the pre-established inclusion criteria. The selected studies were then subjected to a detailed and rigorous review to validate their alignment with the defined inclusion and exclusion criteria. In instances of conflicting opinions among the researchers, the matter was referred to consultation with JS to ensure the selection process remained consistent and coherent.

Relevant data were precisely extracted from the included studies and systematically organized into a structured database, ensuring the quality and consistency of the collected information [35]. Finally, the results were interpreted and synthesized during the analysis phase, critically highlighting potential limitations identified throughout the review process. The final stage entailed formulating a comprehensive systematic review report, encompassing a meticulous exposition of the methodology employed, the results obtained, and the conclusions derived [36,37].

A comprehensive search of specialized databases was conducted to identify pertinent information to support the research. The search strategy was designed to ensure the inclusion of studies related to prediction, prognosis, and diagnosis of dengue using machine learning, deep learning, or artificial intelligence techniques. The specific search syntax applied in each database is detailed in Table 2.

Table 2. Search formula for each database.

IDE	Database	Search Syntax
DB1	Scopus	(TITLE-ABS-KEY ("prediction" OR "prognosis" OR "diagnosis") AND TITLE-ABS-KEY ("dengue") AND TITLE-ABS-KEY ("machine learning" OR "deep learning" OR "artificial intelligence"))
DB2	IEEE Xplore	((("All Metadata": prediction) OR ("All Metadata": diagnosis) OR ("All Metadata": prognosis)) AND ("All Metadata": dengue) AND ((("All Metadata": "machine learning") OR ("All Metadata": "deep learning") OR ("All Metadata": "artificial intelligence"))
DB3	PubMed	((("prognosis"[Mesh] OR "diagnosis"[Mesh] OR "dengue prediction"[tiab:~0]) AND "dengue"[Mesh] AND ("machine learning"[tiab:~0] OR "deep learning"[tiab:~0] OR "artificial intelligence"[tiab:~0]))
DB4	ACM	[[All: prediction] OR [All: prognosis] OR [All: diagnosis]]] AND [[All: dengue] AND [[All: "machine learning"] OR [All: "deep learning"] OR [All: "artificial intelligence"]]]
DB5	ScienceDirect	((Predict OR Diagnosis OR Prognosis) AND Dengue AND ("Machine Learning" OR "Deep Learning" OR "artificial intelligence"))
DB6	Wiley	[[All: predict] OR [All: diagnosis] OR [All: prognosis]]] AND [All: dengue] AND [[All: "machine learning"] OR [All: "deep learning"] OR [All: "artificial intelligence"]]]
DB7	Sage Journals	(Predict OR Diagnosis OR Prognosis) AND Dengue AND ("Machine Learning" OR "Deep Learning" OR "artificial intelligence")

Search terms included controlled terms, such as MesH in PubMed, and free text terms.

2.2. Inclusion and Exclusion Criteria

This protocol rigorously addresses the following aspects: the criteria for study selection, the sources of information used in the literature search, the research strategies implemented, and the procedures for collecting and analyzing the data obtained.

Within the framework of this scientific inquiry, the inclusion and exclusion criteria are defined as pre-established patterns and guidelines that distinguish the studies or articles that will be incorporated into the systematic review from those that will be excluded. Table 3 provides a comprehensive overview of these criteria.

Table 3. Inclusion and exclusion criteria.

Characteristic	Inclusion	Exclusion
Purpose	To predict the early diagnosis of patients with suspected dengue using machine learning and deep learning	Studies using algorithms other than machine learning and deep learning
Participants	Patients with suspected or confirmed dengue fever diagnosis	Patients with other diseases
Time Period	Studies from 2015 to March 2024	Studies outside this time interval
Languages	English	Language other than English

2.3. Sample Selection Process

Following the application of the inclusion and exclusion criteria, a sample selection process was conducted to analyze only articles that provided information relevant to the established research objective. The attached flowchart illustrates that an initial search across six databases identified 2723 articles. After removing duplicates and applying the inclusion and exclusion criteria, 79 articles were retained for eligibility evaluation. Subsequently, additional exclusions were made based on specific reasons outlined during the selection process. The final analysis included 32 articles, as illustrated in Figure 1.

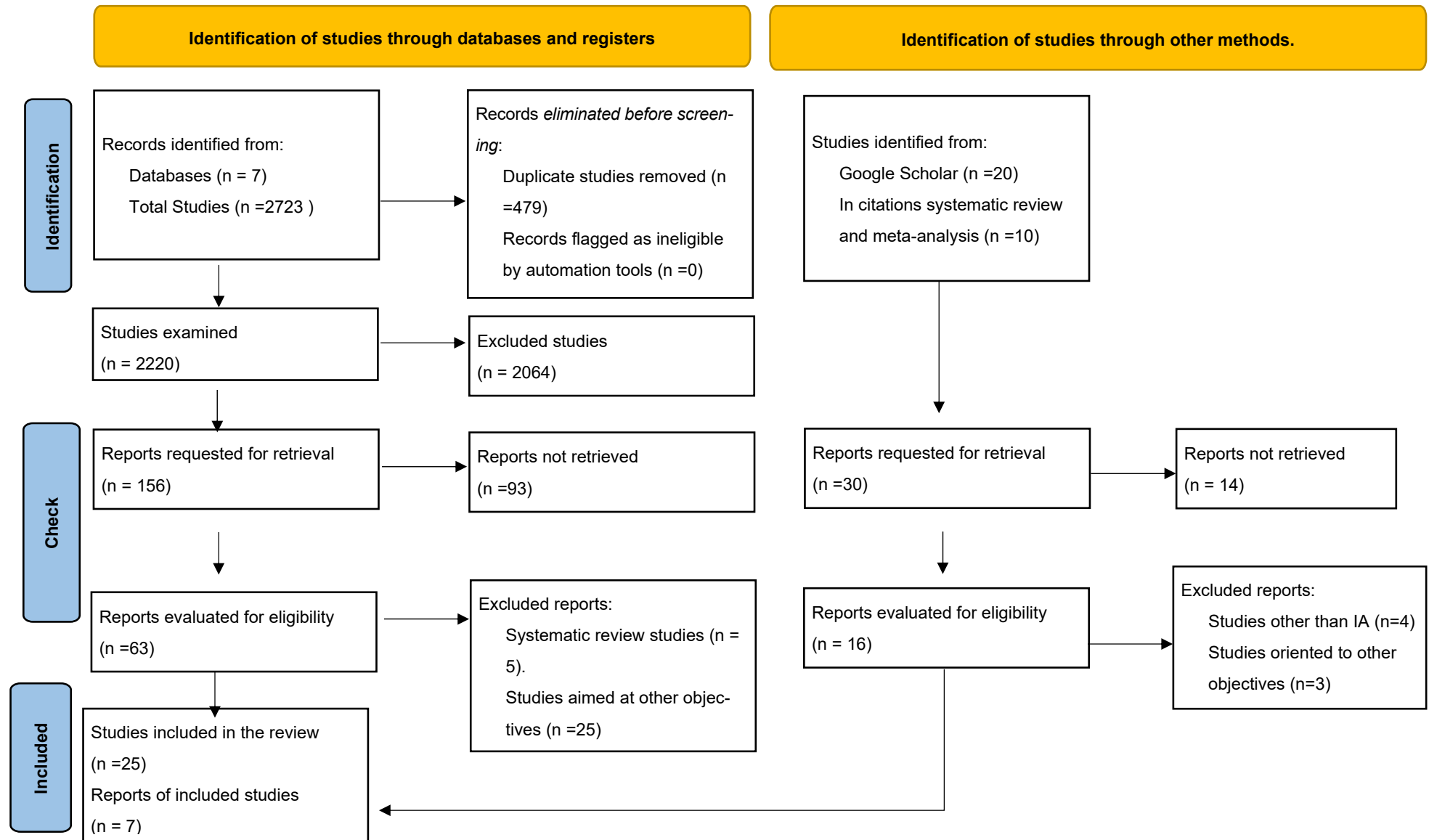


Figure 1. Flow chart of the search method and reference selection for the systematic review.

3. Results

3.1. Study Selection and Characteristics

A total of 2723 studies were identified in seven databases during the preliminary stage of the research. After eliminating 479 duplicate studies and 24 studies for other reasons, 2220 studies remained to be evaluated by title and abstract. Of these, 2064 were excluded, leaving 156 to be retrieved and assessed. However, 93 studies could not be retrieved, making 63 studies available for eligibility determination. Five studies were excluded based on their designation as systematic review studies, twenty-five were excluded for failing to align with the study objective, and eight were excluded for their lack of relevance to artificial intelligence algorithms (see Figure 1).

Regarding identification through alternative methods, only studies that underwent peer review and were published were included. A total of 20 studies were retrieved through Google Scholar, and 10 additional studies were identified from the references of systematic review articles, resulting in 30 studies selected for full-text retrieval. However, 14 studies could not be retrieved, leaving 16 studies for further evaluation and determination of their eligibility. Four studies were excluded because they did not use artificial intelligence algorithms, three because they were focused on other objectives, and two because they were systematic review studies (see Figure 1).

The most salient characteristics in the context of the systematic review are enumerated in Table 4. The following attributes were considered: authorship, year, geographical location, sample size, training strategy, cross-validation, machine learning and deep learning algorithms used, performance metrics, best-performing algorithm, and results obtained.

Table 4. Descriptive characteristics of the included regression and classification algorithm studies.

Author	Country	Sample	Training Strategy		Cross-Validation	Type	Algorithm Used	Performance Metric	Chosen Algorithm	Results
			Train	Test						
Gautam [38]	India	8485	90%	10%	10	Regression	SVM, ANN, KNN, and RF	RMSE, MSE, MAE, and PCC	SVM	Pearson correlation coefficient PCC = 0.71
Kakarla [10]	India	59,801	NA	NA	NA	Regression	VAR, SVR, GBM, and LSTM.	RMSE, MSE, MAE, and R2	LSTM	(Train data: RMSE = 0.453, R = 0.9; test data: RMSE = 0.345, R = 0.9
Huang [31]	Taiwan	798	90%	10%	10	Classification	LR, SVM, RF, GBM, and ANN	AUROC = (0.8324 ± 0.0268)	ANN	Equilibrium accuracy: (0.7523 ± 0.0275)
Nguyen [39]	Vietnam	1,618,767	NA	NA	NA	Regression	CNN, LSTM, and LSTM-ATT	RMSE and MAE	LSTM-ATT	Average of 1.60 for RMSE-based classification and 1.95 for MAE-based classification
Dey [30]	Bangladesh	7000	80%	20%	10	Regression	SVR and MLR	RMSE and MAE	SVR	75% prediction (MAE: 4.95)
Saleh [40]	Malaysia	842	90%	10%	10	Regression	LSTM, and SVR	R2, and MAE	LSTM	R2: 0.75; MAE: 8.76
Appice [41]	Italy	25485	NA	NA	NA	Regression	AutoTiC-NN, VAR, M5, SVM, and ANN	RMSE	AutoTiC-NN	RMSE: 92.83, 96.66, 148.09, 169.18, 232.67, 64.55, 174.45
Gupta [42]	India	1872	77.78%	22.22%	10	Regression	KNN, D.T., R.F., and GNB,	MES	RF	MES: 8.72
Guo [43]	China	NA	NA	NA	NA	Regression	SVR, GBM, NBM, LASSO, and GAM	RMSE and R2	SVR	R ² for Yunnan (R square = 0.976), Guangxi (R = 0.970), Hunan (R = 0.997), Fujian (R = 0.981), and Zhejiang (R = 0.985).
Srivastava [44]	India	182	80%	20%	NA	Regression	Logistic SGD, SGD text, SMO, RF, Adaboost, Naïve Bayes, Random Subspace, Bayes Net, and SVM	F-Means and MCC	RF	F-Means: 0.817; MCC: 0.382
Salim [45]	Malaysia	1300	70%	30%	NA	Ranking	CART, MLP, SVM (LINEAR, POLYNOMIAL, RBF), and Bayes network (TAN)	ROC	SVM	Accuracy = 70%, sensitivity = 14%, specificity = 95%, precision = 56%

Table 4. Cont.

Author	Country	Sample	Training Strategy		Cross-Validation	Type	Algorithm Used	Performance Metric	Chosen Algorithm	Results
			Train	Test						
Abdualgalil [21]	India	6694	70%	30%	10	Classification	KNN, GBC, ETCXGB, ETC, and LightGBM	AUC	ETC	Accuracy: 99.12%, f1-scord: 99.08, precision: 99.18%, recall: 99.12%
Iqbal [46]	India	75	NA	NA	10	Ranking	LogitBoost, LR, RF, ANN, SMO, and KNN	ROC = 0.967	LogitBoost	Sensitivity: 90%; specificity 94%
Saleem [47]	Pakistan	62	51%	49%	10	Ranking	PCA-DA, PCA-SVM, PLS-DA and PLSR	ANOVA	PCA-SVM (poly-5)	Accuracy: 99.52%, sensitivity: 99.75%, and specificity: 99.09%
Santos [48]	Brazil	400,202	80%	20%	10	Ranking	RF, LR, and SVM	NA	RF	Accuracy: 85%
Ansari [49]	India	1148	70%	30%	5	Ranking	Adaboost, Extra Trees, GDBT, KNN, LR, MLP, RF, and XGBoost	AUC	XGBoost	Sensitivity: 0.632, specificity: 0.953, f1-scord: 0.714
Davi [50]	Brazil	102	75.5%	24.5%	10	Ranking	ANN and SVM	ROC	ANN	Accuracy: 86%, sensitivity: 98%, specificity: 51%
Kapoor [51]	India	100	NA	NA	NA	Classification	SVM, RF, and DT	NA	SVM	Accuracy: 100%
Ho [52]	China	4894	NA	NA	10	Classification	DNN, DT, and LR	ROC: 83.75% to 85	LR	Sensitivity: 93.1%
Marandi [53]	Australia	877	70%	30%	10	Ranking	MDL, RF, and LingtGBM	AUC = 0.80	RF	Sensitivity: 54.8%; specificity: 87.9%
Liu [29]	USA	365	NA	NA	10	Ranking	XGBoost and ANN	AUC = 0.891	XGBoost	Sensitivity of 86.4% (95% CI 68.2–100) and specificity of 79.7% (95% CI 75.5–83.9)
Sarma [54]	Bangladesh	209	70%	30%	10	Ranking	DT and RF	ROC and AUC	RF	Accuracy: 79% (Nordin et al., 2020)
Nordin [55]	Malaysia	84	70%	30%	10	Ranking	SVM and RBF	MSE, MAE, and RMSE	SVM	Prediction of 85%, sensitivity of 76%, and specificity of 87%.
Khan [56]	Pakistan	84	NA	NA	10	Ranking	SVM	Confusion Matrix (CM)	SVM	Accuracy of 85%, precision of 90%, sensitivity of 73%, and specificity of 93%

Table 4. Cont.

Author	Country	Sample	Training Strategy		Cross-Validation	Type	Algorithm Used	Performance Metric	Chosen Algorithm	Results
			Train	Test						
Tian [57]	China	583	80%	20%	10	Regression	GLM, SVM, GBM, DT, RF, and XGBoost	MAE, RMSE, and R ²	XGBoost	MAE = 89.12, RMSE = 156.07, and R ² = 0.83
Kuo [58]	Taiwan	805	80%	20%	10	Ranking	RF	AUC: 0.9547	RF	Accuracy: 0.8994, sentimentality: 0.8254, specificity: 0.8254
Kesorn [59]	Taiwan	468	NA	NA	10	Ranking	KNN, DT, NN, and SVM	MC	SVM	Sensitivity of 0.8747 and specificity of 0.8747, prediction accuracy of 96.26%
Widayani [60]	Indonesia	1183	70%	30%	NA	Ranking	SVM	AUC: 0.75 Kappa: 0.7552	SVM	Accuracy: 90%
Chaw [61]	Malaysia	170	70%	30%	NA	Ranking	LR, DT, SVM, SVM, ANN, and AdaBoost	AUC: 0.64	SVM (RBF core)	Accuracy: 0.89, Recall: 0.91, F1-Score: 0.90
Salami [62]	Portugal	1937	70%	30%	10	Ranking	RF and xgboost	AUC: 0.94	XGboost	Sensitivity: 0.79, specificity: 0.93
Hamdani [63]	Indonesia	110	NA	NA	10	Classification	C.45, DT, KNN, RF, and SVM	NA	SVM	Precision: 99.1, recall: 99.1, accuracy: 99.1
Rahmawati [64]	Taiwan	795	70%	30%	4	Rating	C-SVC	NA	C-SVC(RBF)	Accuracy: 98.1%, accuracy: 100%, recall: 100%.

The following methods were identified: General Linear Model (GLM), support vector machine (SVM), Gradient Boosting Machine (GBM) and decision tree (DT), Random Forest (RF), Extreme Gradient Boosting Algorithms (XGBoost), Automatic Vector Regression (VAR), Support Vector Regression (SVR), Long-Term Memory Models (LSTM), logistic regression (LR), Automatic Time-series Classification using Neural Networks (AutoTiC-NN), Neural Networks (ANN), k-nearest neighbors (KNN), Multiple Linear Regression (MLR), Convolutional Neural Network Models (CNN), Long-Term Memory Attention Mechanisms (LSTM-ATT), Mean Shift (MS), Neural Network (NN), Naive Bayes Gaussian (GNB), Naive Bayes Model (NBM), Least Absolute Reduction and Selection Operator (LASSO), Generalized Additive Model (GAM), Minimum Description Length (MDL), efficient and scalable Gradient Boosting Algorithm (LingtGBM), logistic regression using stochastic gradient descent algorithm (Logistic SGD), text classification model using stochastic gradient descent algorithm (SGD text), Sequential Minimal Optimization (SMO), Adaptive Boosting Algorithm (Adaboost), decision trees (CART), Multilayer Artificial Neural Network Perceptron (MLP), radial basis function (RBF), Generalized Linear Model (GLM), Bayes network (TAN), Transformer, Extra Tree, Gradient Boosting Classifier (GBC), Extraordinary Tree Classifier–Extreme Gradient Boosting (ETCXGB), Extremely Random Tree Classifier (ETC), principal component analysis and Discriminant Analysis (PCA-DA), principal component analysis and support vector machine (PCA-SVM), Partial Least Squares Discriminant Analysis (PLS-DA), Partial Least Squares Regression (PLSR), Gradient Boosting Decision Trees (GDBT), machine learning algorithms used for decision tree construction (C4.5), Deep Neural Network (DNN), Support Vector Classification with Parameter C (C-SVC), and the efficient assembly algorithm combining decision trees for binary classification (LogitBoost).

3.2. Risk of Bias Within Studies

According to the risk-of-bias assessment criteria established by QUADAS-2, the results of the retrospective diagnostic accuracy studies are summarized in Table 4. Of the 32 articles subjected to analysis, 7 (21.88%) were evaluated across all domains of the risk-of-bias component as having a low risk of bias. In the patient selection domain, 17 studies (53.13%) were classified as having an unclear risk of bias, primarily due to the lack of a detailed description of the study population, the omission of information regarding data partitioning, and the absence of specification regarding ethical approval.

In the index test domain, 22 studies (68.75%) were considered to have a low risk of bias, consistent with the finding that 23 studies (71.88%) reported adequate model training using cross-validation strategies. In the reference standard domain, 23 studies (71.88%) had a low risk of bias, given that most healthcare centers used RT-PCR (Reverse Transcription–Polymerase Chain Reaction) as the gold standard for dengue diagnosis. Within the flow and timing domain, 29 studies (90.63%) were appraised as having a low risk of bias, as all patients received both the index test and the reference standard, and the time interval between these tests was adjudged to be reasonable.

Concerning the applicability of the concerns component, 17 studies (53.13%) were classified as having a low risk of applicability concerns, as shown in Figures 2 and 3. It is imperative to note that both components, risk of bias and applicability of concerns, play a pivotal role in determining the internal and external validity of the studies analyzed.

	Risk of Bias				Applicability Concerns		
	Patient Selection	Index Test	Reference Standard	Flow and Timing	Patient Selection	Index Test	Reference Standard
(Kapoor, Kadyan, & Ahuja, 2020)	?	+	+	+	?	+	+
Abdualgalil, Abraham, & Ismael, 2022	+	+	+	+	+	+	+
Ansari, Jain, & Budhiraja, 2023	+	+	+	+	+	+	+
Appice, Gel, Iliev, Lyubchich, & Malerba, 2020	+	+	?	?	+	+	?
Chaw, et al. 2024	+	+	?	?	+	?	?
Davi, et al., 2019	?	+	?	+	?	+	?
Dey, et al., 2022	+	+	+	+	+	+	+
Gautam, Thakur, Rajput, & Kumar, 2023	+	+	+	+	+	+	+
Guo, et al., 2017	?	+	+	+	?	+	+
Gupta, et al., 2023	?	+	+	+	?	+	+
Hamdami, et al. 2022	?	+	?	+	+	+	+
Ho, et al., 2020	?	?	+	+	?	+	+
Huang , et al. 2020	?	+	+	+	?	+	+
Iqbal & Islam, 2019	?	+	+	+	?	?	+
Kakarla, et al., 2023	+	?	?	?	+	+	+
Kesorn, et al., 2015	?	+	+	+	?	?	?
Khan, et al., 2016	+	+	?	+	+	+	?
Kuo, Yang, & Su, 2024	?	+	+	+	?	+	+
Liu, et al., 2022	+	?	+	+	+	+	+
Nguyen, et al., 2022	+	+	+	+	?	?	+
Nordin, et al., 2020	?	+	+	+	+	+	+
Rahmawati & Huang, 2016	+	?	+	+	+	+	+
Salami, Sousa, Martins, & Capinha, 2020	?	+	+	+	+	+	+
Saleem, Ali, Bilal, Safdar, & Hassan, 2022	+	+	+	+	+	+	+
Saleh & Baiwei, 2021	?	+	+	+	+	+	+
Salim et al. 2021	?	+	?	+	?	+	+
Santos, Tuboi, & Siqueira, 2023	?	+	+	+	+	+	+
Sarma, et al., 2020	+	+	+	+	+	+	+
Srivastava, Soman, Rai, & Cheema, 2020	+	?	+	+	+	+	+
Tian, et al., 2024	?	+	+	+	+	+	+
Widayani, Sahitya, & Saputri, 2024	?	?	+	+	?	?	+
Zargari, et al., 2023	+	+	+	+	+	+	+

● High ● Unclear ● Low

Figure 2. Methodologic qualities of the included studies assessed according to the Quality Assessment of Diagnostic Accuracy Study 2 tool for risk of bias and applicability concerns. Green represents a low, yellow unclear, and red a high risk of bias.

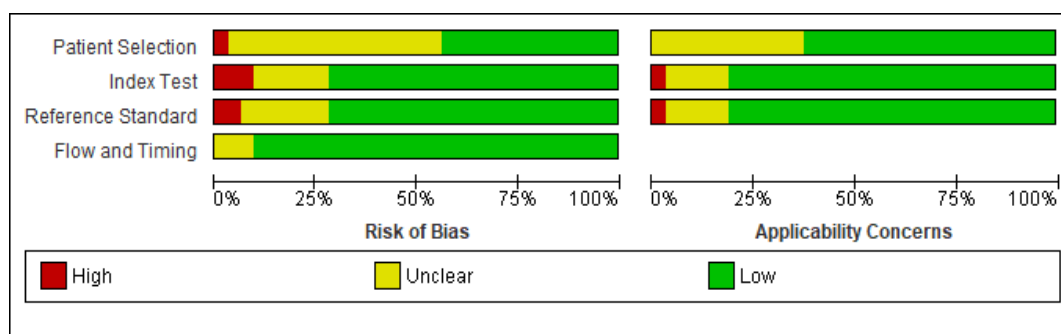


Figure 3. Risk of bias and applicability concerns graph; review authors' judgments about each domain presented as percentages across included studies. Red, yellow, and green frames correspond to high, unclear, and low risk of bias, respectively.

3.3. Results of Data Synthesis

The results of this study reveal the geographical distribution of the articles selected for regression and classification purposes according to their country of origin. Among the 32 articles examined, the preponderance originated from India, accounting for 25% of the total sample size with eight distinct studies. Malaysia and Taiwan each contributed 12.5% (four studies), while China contributed 9.38% (three studies). Bangladesh, Pakistan, Brazil, and Indonesia each contributed two articles, amounting to 6.25% of the total. Additionally, Vietnam, Italy, the USA, Australia, and Portugal contributed one article each, representing 3.13%. These findings suggest that research on dengue prediction and diagnosis has global relevance. However, it is crucial to consider that the choice of database used could have influenced this distribution.

For the distribution of training and test sets, the following schemes were identified in the analyzed studies: 31.25% (10 studies) utilized 70% of the data for training and 30% for testing; 15.63% (5 studies) allocated 80% of the data to training and 20% to testing; and 9.38% (3 studies) assigned 90% of the sample to training and 10% to testing. Furthermore, 9.38% (three studies) utilized non-standard proportions for the training and testing phases. Notably, 34.38% (11 studies) did not explicitly report the proportion used for data splitting. While this may not inherently indicate the absence of data partitioning between training and testing sets, it does compromise the transparency and reproducibility of the process, potentially introducing bias in data selection.

Concerning the cross-validation strategy, it was observed that 65.63% of the studies (21 studies) employed the 10-fold cross-validation method, while 3.13% (1 study) utilized the 5-fold method, and another study applied the 4-fold method. The remaining studies did not explicitly report using k-fold cross-validation, suggesting that alternative validation methods might have been employed or that methodological details were omitted. The lack of uniformity in the validation strategy, compounded in some cases by an absence of methodological transparency, can introduce information bias by affecting the model evaluation during the training and validation process. This, in turn, can compromise the comparability and reproducibility of the results.

Concerning the type of algorithm, 71.88% (23 studies) were classified as classification problems, while 28.12% (9 studies) were classified as regression problems. The evaluation of classification problems employs metrics such as accuracy, precision, sensitivity, specificity, and ROC-AUC, while regression problems are evaluated using metrics like RMSE, MSE, MAE, and PCC.

A combination of 48 machine learning and deep learning algorithms was used for the analysis. The results were distributed as follows: 15 papers used support vector machine (SVM); 13 papers, Random Forest (RF); 8 papers, k-nearest neighbors (K-NN); 6 papers, Artificial Neural Networks (ANN); 6 papers, logistic regression (LR); 8 papers, decision trees (DT); 4 papers, Support Vector Regression (SVR) as well as the Extreme Gradient Boosting Algorithm (XGBoost) and Gradient Boosting Machine (GBM); 7 papers, Multilayer Perceptron (MLP); 3 papers, Long-Term Memory Models (LSTM) and the Adaptive Boosting Algorithm (Adaboost); 2 papers, Vector Automatic Regression (VAR), the efficient and scalable Gradient Boosting Algorithm (LightGBM), and Sequential Minimal Optimization (SMO); and 1 paper used the Bayes network (TAN), a Convolutional Neural Network (CNN), Long-Term Memory with Attention Mechanisms (LSTM-ATT), Multiple Linear Regression (MLR), Gradient Boosting Classifier (GBC), Extraordinary Tree Classifier–Extreme Gradient Boosting (ETCXGB), Extremely Random Tree Classifier (ETC), Classification and Regression Trees (CART), principal component analysis and Discriminant Analysis (PCA-DA), principal component analysis and support vector machine (PCA-SVM), Machine Learning Algorithm for Building Decision Trees (C4.5), Partial Least Squares Discriminant

Analysis (PLS-DA), logistic regression using the Stochastic Gradient Descent Algorithm (Logistic SGD), Text Classifier using the Stochastic Gradient Descent Algorithm (SGD Text), Partial Least Squares Regression (PLSR), Gradient Boosting Decision Trees (GDBT), Boosting Algorithm (LogitBoost), Extra Tree, Mean Shift (MS), Neural Network (NN), Automatic Time-series Classification using Neural Networks (AutoTiC-NN), Gaussian Naive Bayes (GNB), Deep Neural Network (DNN), Naive Bayes Model (NBM), Least Absolute Reduction and Selection Operator (LASSO), Generalized Additive Model (GAM), and Support Vector Classification with the C parameter (C-SVC).

Concerning the machine learning and deep learning algorithms that demonstrated exceptional performance in the reviewed studies, the following findings were obtained: 28.13% of the studies indicated that SVM exhibited the most outstanding performance, followed by Random Forest with 18.75%, and XGBoost, accounting for 12.5% of the total analyzed articles. These results are corroborated by the extant theoretical literature, which documents the superior performance of SVM, Random Forest, and XGBoost in most empirical studies.

The superior performance of SVM, the top-ranked algorithm in 28.13% of the studies, can be attributed to its capacity to address high-dimensional problems and its efficacy in classifying nonlinear data by employing kernels. The robustness of Random Forest, which demonstrated outstanding performance in 18.75% of the studies, stems from its focus on combining multiple decision trees to reduce the risk of overfitting and improve overall accuracy.

These findings are consistent with numerous empirical studies that have demonstrated these algorithms' efficacy in various classification and regression tasks, thereby reinforcing their usefulness and reliability in practical machine learning and deep learning applications.

A geographical analysis of the articles (Figure 4) revealed that 81.25% (26 studies) of the total evaluated articles originate from the Asian continent, while only 6.25% (2 studies) come from South America. Conversely, a mere 3.13% (one study) of the analyzed articles originated from North America, Oceania, and Europe. These findings reveal an evident publication bias, reflected in the lack of uniformity in the distribution of publications across the various continents.

Geographical Distribution of Studies by Continent

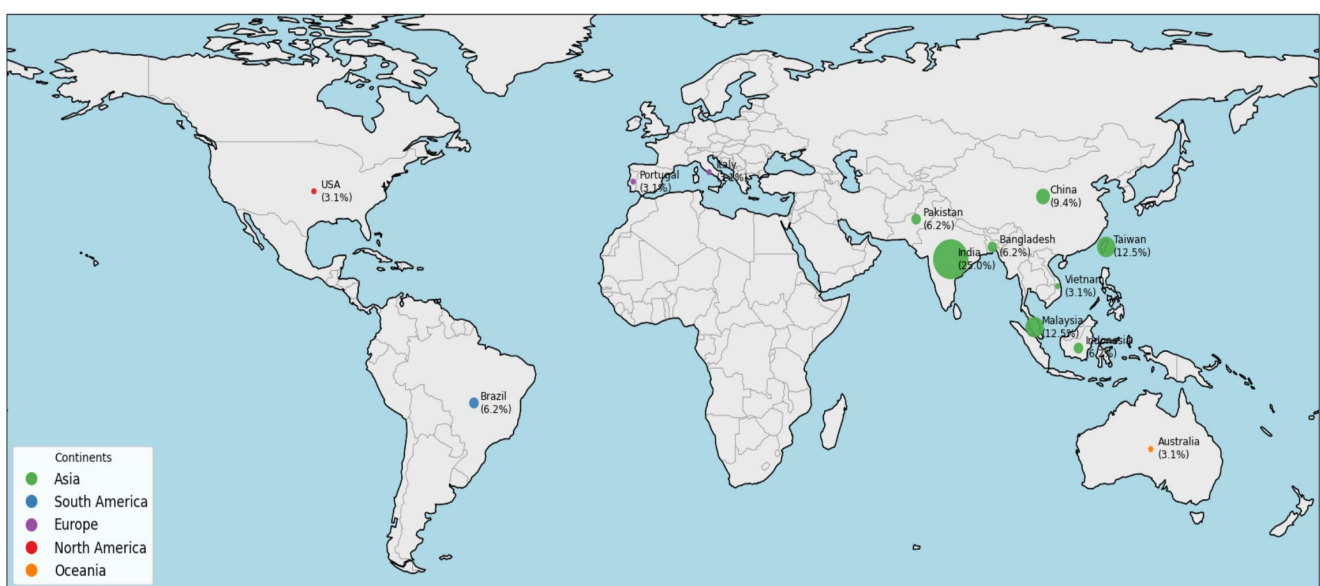


Figure 4. Frequency of articles by country and continent.

4. Discussion

During the document analysis, a broad set of 48 algorithms used in machine learning and deep learning was identified. A higher frequency of outstanding performance was observed when applying SVM and Random Forest algorithms within this wide range. Notably, the model based on the PCA-SVM algorithm (poly-5) demonstrated the most optimal performance, with an accuracy of 99.52%, a sensitivity of 99.75%, and a specificity of 99.09%. This model is a variant of the SVM algorithm. The subsequent discussion will address the theoretical underpinnings that justify the remarkable performance of these algorithms.

SVM is a supervised machine learning algorithm employed for regression and classification tasks. It is well documented that SVM creates multiple hyperplanes, seeking to identify the optimal one with a maximum margin to classify the data as accurately as possible [59,60]. According to [38], SVM demonstrated superior performance in predictive model building compared to RF, kNN, and ANN. These outcomes are consistent with those reported by Salim et al. (2021), who found that SVM (linear kernel) exhibited optimal prediction performance with 70% accuracy, 14% sensitivity, and 95% specificity.

Ref. [55] noted that support vector machines (SVMs) are constructed using three distinct kernel functions: the Gaussian radial basis function (RBF), the polynomial function, and the linear function. This study demonstrated that SVMs with the RBF kernel function outperformed the other techniques, achieving a prediction accuracy of 85%.

In a related study, [56] employed SVM models constructed from three distinct core functions: the Gaussian radial basis function (RBF), the polynomial function, and the linear function. The study revealed that the polynomial kernel of order 1 exhibited the optimal performance, attaining a diagnostic accuracy of approximately 85%, with a precision of 90%, a sensitivity of 73%, and a specificity of 93%.

Improved approaches based on PCA and associated with an SVM have been proposed as highly effective for classifying samples [65]. Furthermore, the application of PCA techniques to reduce the feature dimensions embedded in SVMs for diagnosis and comparison with other diagnostic methods has demonstrated the effectiveness of the proposed approach [66].

Conversely, Random Forest is a learning algorithm based on creating and amalgamating a set of decision trees to obtain a definitive prediction [67]. The construction of each tree within this ensemble is executed independently through the implementation of a technique called bagging [24]. This technique involves the random selection and replacement of subsets of the original training data for the individual construction of decision trees [68].

The theoretical underpinnings of Random Forest's exceptional performance are rooted in its nature as a machine learning ensemble [46]. This concept encompasses methodologies that integrate multiple individual models to enhance the system's predictive capability and resilience [69]. The distinguishing characteristic of Random Forest is its construction of an ensemble of decision trees, wherein each tree represents an individual model within the ensemble [68]. The construction of these trees is achieved by utilizing a random sample, with a replacement drawn from the original training dataset, and through the random selection of attributes at each node [70].

Another salient aspect of Random Forest's efficacy pertains to its capacity to circumvent the predicaments engendered by local optima [70]. Some algorithms, such as decision trees, have the potential to generate highly non-convex cost functions, which may result in the occurrence of local minima points in the methods employed for their resolution [71]. Convergence to local optima can be an obstacle. However, by amalgamating diverse hypotheses through varied approaches in each instance, the likelihood of more accurately approaching the genuine hypothesis is increased [68]. This approach is predicated on

the exploration of multiple solutions and on decreasing the dependence on a single local optimum [71].

The Random Forest algorithm and the SVM model have exhibited remarkable efficacy in numerous applications, particularly in the context of disease diagnosis, as evidenced by their superior performance in predicting diseases such as dengue [42,48,54]. The effectiveness of Random Forest stems from its capacity to address issues related to local optima and overfitting, attributable to the diversity and statistical independence among the trees that constitute the mode [52]. These characteristics render it a particularly robust option in scenarios involving complex and noisy data [72].

However, it is imperative to acknowledge the limitations inherent to each model. In the case of SVMs, their computational complexity, which scales from (n^2) to (n^3) , poses a significant scalability challenge when applied to large datasets [73,74]. Conversely, Random Forests, despite their parallelizable architecture, which renders them more efficient, can also present challenges due to the computational expense associated with training a substantial number of trees, necessitating substantial resources [75].

The efficacy of each algorithm is contingent upon the specific requirements of the given application. SVMs are ideal for data with clear and well-defined separations in moderate-sized datasets, while Random Forests excel in scenarios where the data are noisy or imbalanced. The selection of an algorithm necessitates a judicious balancing of factors, such as accuracy, interpretability, and the availability of computational resources. Ultimately, the optimal choice between these two algorithms depends on the demands and nuances of the problem.

5. Conclusions and Recommendations

This systematic review offers a comprehensive perspective on the early diagnosis of patients with suspected dengue fever, employing machine learning and deep learning techniques. This study identified the most promising algorithms and their respective variants in terms of their predictive ability. The timely diagnosis of patients infected with dengue has significant potential to optimize treatment and significantly reduce mortality in the healthcare setting.

A comprehensive review of 32 scientific papers identified 48 distinct machine learning and deep learning algorithms. The support vector machine (SVM) emerged as the most prevalent algorithm, accounting for 28.13% (nine studies) of the total algorithms analyzed, followed by Random Forest, which accounted for 18.75% (six studies). The model demonstrating the most exceptional performance, attaining 99.52% accuracy, 99.75% sensitivity, and 99.09% specificity, was founded on the PCA-SVM algorithm (poly-5), a variant of the PCA-SVM algorithm.

The Random Forest model is distinguished by its capacity to address the challenges of local optima and overfitting in the machine learning domain. However, to achieve more robust and generalizable results, including relevant variables, conducting additional research addressing larger datasets is essential. Principal component analysis (PCA) has been demonstrated to be a significant performance booster, reducing the dimensionality of the feature space. This finding highlights the efficacy of the PCA-SVM algorithm (poly-5).

These findings underscore the importance of further exploring and refining modeling techniques to maximize the effectiveness of algorithms in predicting and classifying complex data. They also highlight the promising capability of machine learning and deep learning techniques to address the challenge of early diagnosis in patients with suspected dengue.

The results and recommendations presented in this article are expected to be a pivotal starting point for future research and practical applications in the healthcare field.

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References

1. Reinhold, J.M.; Lazzari, C.R.; Lahondère, C. Effects of the Environmental Temperature on *Aedes Aegypti* and *Aedes Albopictus* Mosquitoes: A Review. *Insects* **2018**, *9*, 158. [\[CrossRef\]](#) [\[PubMed\]](#)
2. Tayal, A.; Kabra, S.K.; Lodha, R. Management of Dengue: An Updated Review. *Indian J. Pediatr.* **2023**, *90*, 168–177. [\[CrossRef\]](#) [\[PubMed\]](#)
3. Gebremariam, T.T.; Schallig, H.D.F.H.; Kurmane, Z.M.; Danquah, J.B. Increasing Prevalence of Malaria and Acute Dengue Virus Coinfection in Africa: A Meta-Analysis and Meta-Regression of Cross-Sectional Studies. *Malar. J.* **2023**, *22*, 300. [\[CrossRef\]](#)
4. Zerfu, B.; Kassa, T.; Legesse, M. Epidemiology, Biology, Pathogenesis, Clinical Manifestations, and Diagnosis of Dengue Virus Infection, and Its Trend in Ethiopia: A Comprehensive Literature Review. *Trop. Med. Health* **2023**, *51*, 11. [\[CrossRef\]](#) [\[PubMed\]](#)
5. Lessa, C.L.S.; Hodel, K.V.S.; Gonçalves, M.d.S.; Machado, B.A.S. Dengue as a Disease Threatening Global Health: A Narrative Review Focusing on Latin America and Brazil. *Trop. Med. Infect. Dis.* **2023**, *8*, 241. [\[CrossRef\]](#)
6. Wong, J.M.; Adams, L.E.; Durbin, A.P.; Muñoz-Jordán, J.L.; Poehling, K.A.; Sánchez-González, L.M.; Volkman, H.R.; Paz-Bailey, G. Dengue: A Growing Problem With New Interventions. *Pediatrics* **2022**, *149*, e2021055522. [\[CrossRef\]](#)
7. Malavige, G.N.; Sjö, P.; Singh, K.; Piedagnel, J.-M.; Mowbray, C.; Estani, S.; Lim, S.C.L.; Siquiera, A.M.; Ogg, G.S.; Fraisse, L.; et al. Facing the escalating burden of dengue: Challenges and perspectives. *PLoS Glob. Public Health* **2023**, *3*, e0002598. [\[CrossRef\]](#)
8. Wu, W.; Ren, H.; Lu, L. Increasingly expanded future risk of dengue fever in the Pearl River Delta, China. *PLoS Negl. Trop. Dis.* **2021**, *15*, e0009745. [\[CrossRef\]](#)
9. Sahu, M.C.; Samantaray, R.K.; Pal, A.; Pati, S. Recent advances on pathogenesis, diagnosis, prevention, immunological aspects, and vectors of dengue: A review. *Asian Pac. J. Trop. Biomed.* **2023**, *13*, 325. [\[CrossRef\]](#)
10. Kakarla, S.G.; Kondeti, P.K.; Vavilala, H.P.; Boddada, G.S.B.; Mopuri, R.; Kumaraswamy, S.; Kadiri, M.R.; Mutheneni, S.R. Weather Integrated Multiple Machine Learning Models for Prediction of Dengue Prevalence in India. *Int. J. Biometeorol.* **2023**, *67*, 285–297. [\[CrossRef\]](#)
11. Thomas, S.J.; Yoon, I.-K. A Review of Dengvaxia®: Development to Deployment. *Hum. Human. Vaccines Immunother.* **2019**, *15*, 2295–2314. [\[CrossRef\]](#) [\[PubMed\]](#)
12. Islam, J.; Hu, W. Heatwaves and Dengue Outbreak in Bangladesh After the Pandemic—An Urgent Call for Climate-Driven Early Warning Systems. *Clin. Infect. Dis.* **2024**, *78*, 1075–1076. [\[CrossRef\]](#)
13. Dhoubhadel, B.G.; Hayashi, Y.; Domai, F.M.; Bhattarai, S.; Ariyoshi, K.; Pandey, B.D. A Major Dengue Epidemic in 2022 in Nepal: Need of an Efficient Early-Warning System. *Front. Trop. Dis.* **2023**, *4*, 1217939. [\[CrossRef\]](#)
14. Yang, X.; Quam, M.B.M.; Zhang, T.; Sang, S. Global Burden for Dengue and the Evolving Pattern in the Past 30 Years. *J. Travel Med.* **2021**, *28*, taab146. [\[CrossRef\]](#)
15. Sylvestre, E.; Joachim, C.; Cécilia-Joseph, E.; Bouzillé, G.; Campillo-Gimenez, B.; Cuggia, M.; Cabié, A. Data-driven methods for dengue prediction and surveillance using real-world and Big Data: A systematic review. *PLoS Negl. Trop. Dis.* **2022**, *16*, e0010056. [\[CrossRef\]](#)
16. Tsheten, T.; Gray, D.J.; Clements, A.C.A.; Wangdi, K. Epidemiology and Challenges of Dengue Surveillance in the WHO South-East Asia Region. *Trans. R. Soc. Trop. Med. Hyg.* **2021**, *115*, 583–599. [\[CrossRef\]](#)
17. Cardenas, R.; Hussain-Alkhateeb, L.; Benitez-Valladares, D.; Sánchez-Tejeda, G.; Kroeger, A. The Early Warning and Response System (EWARS-TDR) for Dengue Outbreaks: Can It Also Be Applied to Chikungunya and Zika Outbreak Warning? *BMC Infect. Dis.* **2022**, *22*, 235. [\[CrossRef\]](#)

18. Hussain-Alkhateeb, L.; Kroeger, A.; Olliaro, P.; Rocklöv, J.; Sewe, M.O.; Tejada, G.; Benitez, D.; Gill, B.; Hakim, S.L.; Carvalho, R.G.; et al. Early warning and response system (EWARS) for dengue outbreaks: Recent advancements towards widespread applications in critical settings. *PLoS ONE* **2018**, *13*, e0196811. [\[CrossRef\]](#)
19. Benitez-Valladares, D.; Kroeger, A.; Tejada, G.S.; Hussain-Alkhateeb, L. Validation of the Early Warning and Response System (EWARS) for dengue outbreaks: Evidence from the national vector control program in Mexico. *PLoS Negl. Trop. Dis.* **2021**, *15*, e0009261. [\[CrossRef\]](#)
20. Manogaran, G.; Lopez, D. A Gaussian Process Based Big Data Processing Framework in Cluster Computing Environment. *Clust. Comput.* **2018**, *21*, 189–204. [\[CrossRef\]](#)
21. Abdualgalil, B.; Abraham, S.; Ismael, W.M. Early Diagnosis for Dengue Disease Prediction Using Efficient Machine Learning Techniques Based on Clinical Data. *J. Robot. Control* **2022**, *3*, 257–268. [\[CrossRef\]](#)
22. Miranda, S.A.D.; Aguilar, R.R. Machine Learning Models in Health Prevention and Promotion and Labor Productivity: A Co-Word Analysis. *Iberoam. J. Sci. Meas. Commun.* **2024**, *4*, 1–16. [\[CrossRef\]](#)
23. Runge-Ranzinger, S.; McCall, P.J.; Kroeger, A.; Horstick, O. Dengue disease surveillance: An updated systematic literature review. *Trop. Med. Int. Health* **2014**, *19*, 1116–1160. [\[CrossRef\]](#)
24. da Silveira, L.T.C.; Tura, B.; Santos, M. Systematic Review of Dengue Vaccine Efficacy. *BMC Infect. Dis.* **2019**, *19*, 750. [\[CrossRef\]](#) [\[PubMed\]](#)
25. Ahmed, A.M.; Mohammed, A.T.; Vu, T.T.; Khattab, M.; Doheim, M.F.; Ashraf Mohamed, A.; Abdelhamed, M.M.; Shamandy, B.E.; Dawod, M.T.; Alesaei, W.A.; et al. Prevalence and Burden of Dengue Infection in Europe: A Systematic Review and Meta-Analysis. *Rev. Med. Virol.* **2020**, *30*, e2093. [\[CrossRef\]](#)
26. Dao Phuoc, T.; Khuong Quynh, L.; Vien Dang Khanh, L.; Ong Phuc, T.; Le Sy, H.; Le Ngoc, T.; Phung Khanh, L. Clinical Prognostic Models for Severe Dengue: A Systematic Review Protocol. *Wellcome Open Res.* **2019**, *4*, 12. [\[CrossRef\]](#)
27. Aswi, A.; Cramb, S.M.; Moraga, P.; Mengersen, K. Bayesian Spatial and Spatio-Temporal Approaches to Modelling Dengue Fever: A Systematic Review. *Epidemiol. Infect.* **2019**, *147*, e33. [\[CrossRef\]](#)
28. Hoyos, W.; Aguilar, J.; Toro, M. Dengue Models Based on Machine Learning Techniques: A Systematic Literature Review. *Artif. Intell. Med.* **2021**, *119*, 102157. [\[CrossRef\]](#)
29. Liu, Y.E.; Saul, S.; Rao, A.M.; Robinson, M.L.; Agudelo Rojas, O.L.; Sanz, A.M.; Verghese, M.; Solis, D.; Sibai, M.; Huang, C.H.; et al. An 8-Gene Machine Learning Model Improves Clinical Prediction of Severe Dengue Progression. *Genome Med.* **2022**, *14*, 33. [\[CrossRef\]](#)
30. Dey, S.K.; Rahman, M.M.; Howlader, A.; Siddiqi, U.R.; Uddin, K.M.M.; Borhan, R.; Rahman, E.U. Prediction of dengue incidents using hospitalized patients, metrological and socio-economic data in Bangladesh: A machine learning approach. *PLoS ONE* **2022**, *17*, e0270933. [\[CrossRef\]](#)
31. Huang, S.-W.; Tsai, H.-P.; Hung, S.-J.; Ko, W.-C.; Wang, J.-R. Assessing the risk of dengue severity using demographic information and laboratory test results with machine learning. *PLoS Negl. Trop. Dis.* **2020**, *14*, e0008960. [\[CrossRef\]](#) [\[PubMed\]](#)
32. Yepes-Nuñez, J.J.; Urrútia, G.; Romero-García, M.; Alonso-Fernández, S. The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews Declaración PRISMA 2020: Una Guía Actualizada Para La Publicación de Revisiones Sistemáticas. *Rev. Española De Cardiol.* **2021**, *74*, 790–799. [\[CrossRef\]](#)
33. Sánchez Serrano, S.; Pedraza Navarro, I.; Donoso González, M. ¿Cómo hacer una revisión sistemática siguiendo el protocolo PRISMA?: Usos y estrategias fundamentales para su aplicación en el ámbito educativo a través de un caso práctico. *Bordón Rev. Pedagog.* **2022**, *74*, 51–66. [\[CrossRef\]](#)
34. Schwarzer, G.; Carpenter, J.R.; Rücker, G. *Meta-Analysis with R; Use R!* Springer International Publishing: Cham, Switzerland, 2015; ISBN 978-3-319-21415-3.
35. Andrade-Girón, D.; Marín-Rodríguez, W.; Sandivar-Rosas, J.; Carreño-Cisneros, E.; Susanibar-Ramirez, E.; Zuñiga-Rojas, M.; Angeles-Morales, J.; Villarreal-Torres, H. Generative Artificial Intelligence in Higher Education Learning: A Review Based on Academic Databases. *Iberoam. J. Sci. Meas. Commun.* **2024**, *4*, 1–16. [\[CrossRef\]](#)
36. Alexander, P.A. Methodological Guidance Paper: The Art and Science of Quality Systematic Reviews. *Rev. Educ. Res.* **2020**, *90*, 6–23. [\[CrossRef\]](#)
37. Pigott, T.D.; Polanin, J.R. Methodological Guidance Paper: High-Quality Meta-Analysis in a Systematic Review. *Rev. Educ. Res.* **2020**, *90*, 24–46. [\[CrossRef\]](#)
38. Gautam, S.; Thakur, A.; Rajput, A.; Kumar, M. Anti-Dengue: A Machine Learning-Assisted Prediction of Small Molecule Antivirals against Dengue Virus and Implications in Drug Repurposing. *Viruses* **2024**, *16*, 45. [\[CrossRef\]](#)
39. Nguyen, V.-H.; Tuyet-Hanh, T.T.; Mulhall, J.; Minh, H.V.; Duong, T.Q.; Chien, N.V.; Nhung, N.T.T.; Lan, V.H.; Minh, H.B.; Cuong, D.; et al. Deep learning models for forecasting dengue fever based on climate data in Vietnam. *PLoS Negl. Trop. Dis.* **2022**, *16*, e0010509. [\[CrossRef\]](#)

40. Saleh, A.Y.; Baiwei, L. Dengue Prediction Using Deep Learning With Long Short-Term Memory. In Proceedings of the 2021 1st International Conference on Emerging Smart Technologies and Applications (eSmarTA), Sana'a, Yemen, 10–12 August 2021; pp. 1–5. [\[CrossRef\]](#)
41. Appice, A.; Gel, Y.R.; Iliev, I.; Lyubchich, V.; Malerba, D. A Multi-Stage Machine Learning Approach to Predict Dengue Incidence: A Case Study in Mexico. *IEEE Access* **2020**, *8*, 52713–52725. [\[CrossRef\]](#)
42. Gupta, G.; Khan, S.; Guleria, V.; Almjally, A.; Alabdullah, B.I.; Siddiqui, T.; Albahlal, B.M.; Alajlan, S.A.; AL-subaie, M. DDPM: A Dengue Disease Prediction and Diagnosis Model Using Sentiment Analysis and Machine Learning Algorithms. *Diagnostics* **2023**, *13*, 1093. [\[CrossRef\]](#)
43. Guo, P.; Liu, T.; Zhang, Q.; Wang, L.; Xiao, J.; Zhang, Q.; Luo, G.; Li, Z.; He, J.; Zhang, Y.; et al. Developing a dengue forecast model using machine learning: A case study in China. *PLoS Negl. Trop. Dis.* **2017**, *11*, e0005973. [\[CrossRef\]](#) [\[PubMed\]](#)
44. Srivastava, S.; Soman, S.; Rai, A.; Cheema, A.S. An Online Learning Approach for Dengue Fever Classification. In Proceedings of the 2020 IEEE 33rd International Symposium on Computer-Based Medical Systems (CBMS), Rochester, MN, USA, 28–30 July 2020; pp. 163–168. [\[CrossRef\]](#)
45. Salim, N.A.M.; Wah, Y.B.; Reeves, C.; Smith, M.; Yaacob, W.F.W.; Mudin, R.N.; Dapari, R.; Sapri, N.N.F.F.; Haque, U. Prediction of Dengue Outbreak in Selangor Malaysia Using Machine Learning Techniques. *Sci. Rep.* **2021**, *11*, 939. [\[CrossRef\]](#) [\[PubMed\]](#)
46. Iqbal, N.; Islam, M. Machine Learning for Dengue Outbreak Prediction: A Performance Evaluation of Different Prominent Classifiers. *Informatica* **2019**, *43*, 363–371. [\[CrossRef\]](#)
47. Saleem, M.; Ali, S.; Bilal, M.; Safdar, K.; Hassan, M. Development of Multivariate Classification Models for the Diagnosis of Dengue Virus Infection. *Photodiagnosis Photodyn. Ther.* **2022**, *40*, 103136. [\[CrossRef\]](#)
48. Santos, C.Y.; Tuboi, S.; Abreu, A. de J.L. de; Abud, D.A.; Neto, A.A.L.; Pereira, R.; Siqueira, J.B. A Machine Learning Model to Assess Potential Misdiagnosed Dengue Hospitalization. *Heliyon* **2023**, *9*, e16634. [\[CrossRef\]](#)
49. Ansari, M.S.; Jain, D.; Budhiraja, S. Machine-Learning Prediction Models for Any Blood Component Transfusion in Hospitalized Dengue Patients. *Hematol. Transfus. Cell Ther.* **2024**, *46*, S13–S23. [\[CrossRef\]](#)
50. Davi, C.; Pastor, A.; Oliveira, T.; Neto, F.B.d.L.; Braga-Neto, U.; Bigham, A.W.; Bamshad, M.; Marques, E.T.A.; Acioli-Santos, B. Severe Dengue Prognosis Using Human Genome Data and Machine Learning. *IEEE Trans. Biomed. Eng.* **2019**, *66*, 2861–2868. [\[CrossRef\]](#)
51. Kapoor, R.; Kadyan, V.; Ahuja, S. Weight Based-Artificial Neural Network (W-Ann) For Predicting Dengue Using Machine Learning Approach with Indian Perspective. *Int. J. Sci. Technol. Res.* **2020**, *9*, 3290–3298.
52. Ho, T.-S.; Weng, T.-C.; Wang, J.-D.; Han, H.-C.; Cheng, H.-C.; Yang, C.-C.; Yu, C.-H.; Liu, Y.-J.; Hu, C.H.; Huang, C.-Y.; et al. Comparing machine learning with case-control models to identify confirmed dengue cases. *PLoS Negl. Trop. Dis.* **2020**, *14*, e0008843. [\[CrossRef\]](#)
53. Marandi, R.Z.; Leung, P.; Sigera, C.; Murray, D.D.; Weeratunga, P.; Fernando, D.; Rodrigo, C.; Rajapakse, S.; MacPherson, C.R. Development of a machine learning model for early prediction of plasma leakage in suspected dengue patients. *PLoS Negl. Trop. Dis.* **2023**, *17*, e0010758. [\[CrossRef\]](#)
54. Sarma, D.; Hossain, S.; Mittra, T.; Bhuiya, M.A.M.; Saha, I.; Chakma, R. Dengue Prediction Using Machine Learning Algorithms. In Proceedings of the 2020 IEEE 8th R10 Humanitarian Technology Conference (R10-HTC), Kuching, Malaysia, 1–3 December 2020; pp. 1–6. [\[CrossRef\]](#)
55. Nordin, N.I.; Mohd Sobri, N.; Ismail, N.A.; Zulkifli, S.N.; Abd Razak, N.F.; Mahmud, M. The Classification Performance Using Support Vector Machine for Endemic Dengue Cases. *J. Phys. Conf. Ser.* **2020**, *1496*, 012006. [\[CrossRef\]](#)
56. Khan, S.; Ullah, R.; Khan, A.; Wahab, N.; Bilal, M.; Ahmed, M. Analysis of Dengue Infection Based on Raman Spectroscopy and Support Vector Machine (SVM). *Biomed. Opt. Express BOE* **2016**, *7*, 2249–2256. [\[CrossRef\]](#) [\[PubMed\]](#)
57. Tian, N.; Zheng, J.-X.; Li, L.-H.; Xue, J.-B.; Xia, S.; Lv, S.; Zhou, X.-N. Precision Prediction for Dengue Fever in Singapore: A Machine Learning Approach Incorporating Meteorological Data. *Trop. Med. Infect. Dis.* **2024**, *9*, 72. [\[CrossRef\]](#) [\[PubMed\]](#)
58. Kuo, C.-Y.; Yang, W.-W.; Su, E.C.-Y. Improving Dengue Fever Predictions in Taiwan Based on Feature Selection and Random Forests. *BMC Infect. Dis.* **2024**, *24*, 334. [\[CrossRef\]](#) [\[PubMed\]](#)
59. Kesorn, K.; Ongeluk, P.; Chompoonsri, J.; Phumee, A.; Thavara, U.; Tawatsin, A.; Siriyasatien, P. Morbidity Rate Prediction of Dengue Hemorrhagic Fever (DHF) Using the Support Vector Machine and the Aedes aegypti Infection Rate in Similar Climates and Geographical Areas. *PLoS ONE* **2015**, *10*, e0125049. [\[CrossRef\]](#)
60. Widayani, P.; Sahitya, A.F.; Saputri, A.A. Support Vector Machine for Susceptibility Modeling of Dengue Fever in Kendari, Southeast Sulawesi. *Geomat. Environ. Eng.* **2024**, *18*, 29–50. [\[CrossRef\]](#)
61. Chaw, J.K.; Chaw, S.H.; Quah, C.H.; Sahrani, S.; Ang, M.C.; Zhao, Y.; Ting, T.T. A Predictive Analytics Model Using Machine Learning Algorithms to Estimate the Risk of Shock Development among Dengue Patients. *Healthc. Anal.* **2024**, *5*, 100290. [\[CrossRef\]](#)
62. Salami, D.; Sousa, C.A.; Martins, M.d.R.O.; Capinha, C. Predicting Dengue Importation into Europe, Using Machine Learning and Model-Agnostic Methods. *Sci. Rep.* **2020**, *10*, 9689. [\[CrossRef\]](#)

63. Hamdani, H.; Hatta, H.R.; Puspitasari, N.; Septiarini, A.; Henderi, H. Dengue Classification Method Using Support Vector Machines and Cross-Validation Techniques. *IAES Int. J. Artif. Intell. (IJ-AI)* **2022**, *11*, 1119–1129. [[CrossRef](#)]
64. Rahmawati, D.; Huang, Y.-P. Using C-Support Vector Classification to Forecast Dengue Fever Epidemics in Taiwan. In Proceedings of the 2016 International Conference on System Science and Engineering (ICSSE), Puli, Taiwan, 7–9 July 2016; pp. 1–4.
65. Jin, Y.; Liu, B.; Li, C.; Shi, S. Origin identification of *Cornus officinalis* based on PCA-SVM combined model. *PLoS ONE* **2023**, *18*, e0282429. [[CrossRef](#)]
66. Sucharita, S.; Sahu, B.; Swarnkar, T. An Empirical Analysis of PCA-SVM Model for Cancer Microarray Data Classification. In *Advances in Intelligent Computing and Communication*; Das, S., Mohanty, M.N., Eds.; Springer: Singapore, 2021; pp. 495–504.
67. La Salvia, M.; Torti, E.; Secco, G.; Bellazzi, C.; Salinaro, F.; Lago, P.; Danese, G.; Perlini, S.; Leporati, F. Machine-Learning-Based COVID-19 and Dyspnoea Prediction Systems for the Emergency Department. *Appl. Sci.* **2022**, *12*, 10869. [[CrossRef](#)]
68. Géron, A. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*; O'Reilly Media, Inc.: Sebastopol, CA, USA, 2022; ISBN 978-1-0981-2246-1.
69. Xiao, T.; Zhu, J.; Liu, T. Bagging and Boosting Statistical Machine Translation Systems. *Artif. Intell.* **2013**, *195*, 496–527. [[CrossRef](#)]
70. Adetunji, A.B.; Akande, O.N.; Ajala, F.A.; Oyewo, O.; Akande, Y.F.; Oluwadara, G. House Price Prediction Using Random Forest Machine Learning Technique. *Procedia Comput. Sci.* **2022**, *199*, 806–813. [[CrossRef](#)]
71. Yaman, M.A.; Rattay, F.; Subasi, A. Comparison of Bagging and Boosting Ensemble Machine Learning Methods for Face Recognition. *Procedia Comput. Sci.* **2021**, *194*, 202–209. [[CrossRef](#)]
72. Purkayastha, S.; Xiao, Y.; Jiao, Z.; Thepumnoeysuk, R.; Halsey, K.; Wu, J.; Tran, T.M.L.; Hsieh, B.; Choi, J.W.; Wang, D.; et al. Machine Learning-Based Prediction of COVID-19 Severity and Progression to Critical Illness Using CT Imaging and Clinical Data. *Korean J. Radiol.* **2021**, *22*, 1213–1224. [[CrossRef](#)]
73. Ray, S. An Analysis of Computational Complexity and Accuracy of Two Supervised Machine Learning Algorithms—K-Nearest Neighbor and Support Vector Machine. In *Data Management, Analytics and Innovation*; Sharma, N., Chakrabarti, A., Balas, V.E., Martinovic, J., Eds.; Springer: Singapore, 2021; pp. 335–347.
74. Wang, H.; Zhu, Z.; Shao, Y. Fast Support Vector Machine With Low-Computational Complexity for Large-Scale Classification. *IEEE Trans. Syst. Man. Cybern. Syst.* **2024**, *54*, 4151–4163. [[CrossRef](#)]
75. Zhu, T. Analysis on the Applicability of the Random Forest. *J. Phys. Conf. Ser.* **2020**, *1607*, 012123. [[CrossRef](#)]

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